AWS CERTIFIED CLOUD PRACTITIONER • CLF-C02

Mathias AWS Cloud Practitioner Revision

This page is rebuilt around the official AWS Cloud Practitioner exam guide, the official Technologies and Concepts list, and the official In-Scope AWS Services list. The domain principles come directly from the CLF-C02 blueprint, and the service section follows AWS's own exam categories.

EXAM BLUEPRINT

Official Principles and Core Ideas

These topic blocks are based directly on the four official CLF-C02 domain pages. Each task includes a short study description plus the main knowledge and skill points AWS calls out.

Content Domain 1: Cloud Concepts

Task Statement 1.1: Define the benefits of the AWS Cloud

The **value proposition** of AWS connects cloud adoption with **global infrastructure**, **speed of deployment**, **global reach**, **high availability**, **elasticity**, and **agility**.

Task Statement 1.2: Identify design principles of the AWS Cloud

The **AWS Well-Architected Framework** is organized around six pillars: **Operational Excellence**, **Security**, **Reliability**, **Performance Efficiency**, **Cost Optimization**, and **Sustainability**. Each pillar highlights a different design priority and together they form AWS's main architecture reference model.

Task Statement 1.3: Understand the benefits of and strategies for migration to the AWS Cloud

Cloud adoption strategies and **resources that support migration** include the **AWS Cloud Adoption Framework (AWS CAF)**, which organizes cloud adoption across business, people, governance, platform, security, and operations perspectives. Migration approaches include **database replication** and offline transfer with **AWS Snowball**.

Task Statement 1.4: Understand concepts of cloud economics

Cloud economics includes **fixed versus variable costs**, **on-premises cost considerations**, **licensing strategies** such as Bring Your Own License (BYOL) versus included licenses, **rightsizing**, **benefits of automation**, and **economies of scale**.

Content Domain 2: Security and Compliance

Task Statement 2.1: Understand the AWS shared responsibility model

The **shared responsibility model** divides security into **AWS responsibilities** for the cloud and **customer responsibilities** in the cloud. Customers manage items like identities, data, and configuration, while AWS manages the underlying infrastructure. The balance changes with services such as **Amazon RDS**, **AWS Lambda**, and **Amazon EC2**.

Task Statement 2.2: Understand AWS Cloud security, governance, and compliance concepts

Compliance and governance concepts, **benefits of cloud security** such as encryption, and **where security logs are captured** form the core of this task. Key ideas include where compliance information is published, how customers secure resources, what **encryption in transit** and **encryption at rest** mean, and which services support monitoring, auditing, governance, and reporting.

Task Statement 2.3: Identify AWS access management capabilities

Identity and access management includes **AWS IAM**, the **importance of protecting the root user**, the **principle of least privilege**, and **AWS IAM Identity Center**. Related points include access keys, password policies, secrets storage, multi-factor authentication (MFA), cross-account roles, groups, custom and managed policies, root-user-only tasks, and federated identity concepts.

Task Statement 2.4: Identify components and resources for security

Security capabilities and **security documentation** provided by AWS include services such as **AWS WAF**, **AWS Firewall Manager**, **AWS Shield**, **Amazon GuardDuty**, and **AWS Trusted Advisor**, along with security information published through the Knowledge Center and AWS documentation sources.

Content Domain 3: Cloud Technology and Services

Task Statement 3.1: Define methods of deploying and operating in the AWS Cloud

Ways of provisioning and operating in AWS, **ways to access AWS services**, and **deployment models** include the **AWS Management Console**, **APIs**, **software development kits (SDKs)**, **AWS CLI**, and **infrastructure as code (IaC)**, along with cloud, hybrid, and on-premises deployment models.

Task Statement 3.2: Define the AWS global infrastructure

This task includes **Regions**, **Availability Zones**, and **edge locations**. A **Region** is a separate geographic area, an **Availability Zone** is an isolated location inside a Region, and an **edge location** helps deliver content closer to users. Together they support **high availability**, **low latency**, **disaster recovery**, and **data sovereignty**.

Task Statement 3.3: Identify AWS compute services

Compute services include suitable **Amazon EC2 instance types**, **container services** such as Amazon ECS and Amazon EKS, **serverless compute** such as AWS Fargate and AWS Lambda, **Auto Scaling** for elasticity, and **load balancers** for traffic distribution.

Task Statement 3.4: Identify AWS database services

Database services and **database migration** include the distinction between **EC2-hosted databases** and **managed databases**. **SQL databases** include Amazon RDS and Amazon Aurora. **NoSQL databases** include Amazon DynamoDB. **Memory-based databases** include Amazon ElastiCache, and migration tools include **AWS DMS** and **AWS SCT**.

Task Statement 3.5: Identify AWS network services

Network services include **virtual private cloud (VPC)** components such as subnets and gateways, VPC security concepts such as **network ACLs** and **security groups**, the role of **Amazon Route 53**, and connectivity options such as **AWS VPN** and **AWS Direct Connect**.

Task Statement 3.6: Identify AWS storage services

Storage services include use cases for **object storage**, differences among **Amazon S3 storage classes**, **block storage** such as Amazon EBS and instance store, **file services** such as Amazon EFS and Amazon FSx, cached and hybrid storage options such as **AWS Storage Gateway**, along with **lifecycle policies** and **AWS Backup**.

Task Statement 3.7: Identify AWS AI/ML services and analytics services

AI/ML services and **analytics services** include the tasks performed by services such as **Amazon SageMaker AI**, **Amazon Lex**, and **Amazon Kendra**, along with analytics services such as **Amazon Athena**, **Amazon Kinesis**, **AWS Glue**, and **Amazon QuickSight**.

Task Statement 3.8: Identify services from other in-scope AWS service categories

This task includes additional in-scope categories such as **application integration**, **business applications**, **customer enablement**, **developer tools**, **end-user computing**, **frontend web and mobile**, and **Internet of Things (IoT)**. The core idea is matching the right service to messaging, alerts, support, development, frontend delivery, end-user access, and IoT device management.

Content Domain 4: Billing, Pricing, and Support

Task Statement 4.1: Compare AWS pricing models

Compute purchasing options and **storage pricing options** include **On-Demand Instances** for short-term flexibility, **Reserved Instances** for steady workloads, **Spot Instances** for interruption-tolerant workloads, **AWS Savings Plans** for discounted usage commitments, **Dedicated Hosts** and **Dedicated Instances** for isolation needs, and **Capacity Reservations** for guaranteed EC2 capacity. Related pricing points include **Reserved Instance flexibility**, how Reserved Instances behave in **AWS Organizations**, data transfer pricing, and differences among storage tiers.

Task Statement 4.2: Understand resources for billing, budget, and cost management

Billing information, **pricing information**, **AWS Organizations**, and **cost allocation tags** form the core of this task. Important tools include **AWS Budgets**, **AWS Cost Explorer**, and **AWS Pricing Calculator**, along with consolidated billing and the role of cost allocation tags in reports such as the **AWS Cost and Usage Report**.

Task Statement 4.3: Identify AWS technical resources and AWS Support options

Official AWS documentation and resources, **AWS Support plans**, **AWS Support Center**, the **AWS Partner Network**, and AWS experts define this task. Relevant resources include **whitepapers**, **blogs**, **documentation**, **AWS Prescriptive Guidance**, **AWS Knowledge Center**, and **AWS re:Post**. Support options include **Developer Support** with **technical guidance for development and test**, **Business Support** with **24/7 access to AWS experts** and the **full set of AWS Trusted Advisor checks**, **Enterprise On-Ramp Support** with a **Technical Account Manager (TAM)** for growing business-critical workloads, and **Enterprise Support** with a **designated TAM** and the most **proactive guidance** for enterprise environments. Current AWS Support pages emphasize **account and billing support**, **AWS Trusted Advisor**, and **TAM guidance** as key plan differences. This task also includes the roles of **AWS Support**, **AWS Trusted Advisor**, **AWS Health Dashboard**, **AWS Health API**, **AWS Marketplace**, **AWS Professional Services**, and **AWS Solutions Architects**.

HIGH-YIELD TOPICS

Official Technologies and Concepts

Architecture and Build

- **APIs** : programmatic interfaces for AWS operations.
- **Infrastructure as code (IaC)** : define infrastructure with templates.
- **AWS SDKs** : developer libraries for AWS services.
- **AWS Well-Architected Framework** : best-practice design guidance.
- **Compute** : processing workloads in the cloud.
- **Databases** : managed and self-managed data stores.
- **Storage** : object, block, file, and archive storage.
- **Network services** : connectivity, routing, and delivery services.

Cloud and Migration

- **Benefits of migrating to the AWS Cloud** : gain agility, scalability, and reach.
- **AWS Cloud Adoption Framework (AWS CAF)** : structure cloud transformation planning.
- **Migration and data transfer** : move servers, databases, and files to AWS.
- **AWS global infrastructure** : Regions and Availability Zones support resiliency.
- **Amazon EC2 instance types** : match compute size and pricing to workloads.
- **Machine learning** : use AI/ML services for predictions and automation.
- **Management and governance** : monitor, control, and govern AWS resources.

Security and Cost

- **Security** : protect identities, data, and workloads.
- **AWS shared responsibility model** : AWS secures the cloud; customers secure what they run.
- **AWS Compliance** : compliance programs and audit resources.
- **AWS Security Blog** : official AWS security updates and guidance.
- **Cost management** : monitor and optimize cloud spending.
- **AWS Pricing Calculator** : estimate cost before deployment.

Support and Guidance

- **AWS Knowledge Center** : official troubleshooting articles.
- **AWS Prescriptive Guidance** : strategies, patterns, and implementation guidance.
- **AWS re:Post** : community and AWS knowledge Q&A.
- **AWS Support Center** : place to manage support cases.
- **AWS Support plans** : **Business** adds **24/7 expert access** and **AWS Trusted Advisor**, **Enterprise On-Ramp** adds a **Technical Account Manager (TAM)**, and **Enterprise** provides the deepest **proactive TAM guidance**.
- **AWS Partner Network (APN)** : AWS partner ecosystem.
- **AWS Professional Services** : AWS expert delivery support.
- **AWS solutions architects** : AWS design and architecture guidance.

REVISION FOCUS

Exam Topics and Common Question Patterns

Based on the official CLF-C02 scope and the patterns that appear frequently in public practice-question sets, these are the topics that tend to come up again and again in revision and mock exams.

High-Frequency Topics

- **Shared responsibility model** : **AWS secures the cloud**; the **customer secures what runs in the cloud**, such as identities, data, operating systems on Amazon EC2, and configurations.
- **IAM, MFA, and root user** : the best answer is usually **use least privilege**, **enable MFA**, and **do not use the root user for everyday tasks**.
- **Regions and Availability Zones** : a **Region** is a geographic area, an **Availability Zone** is an isolated location inside a Region, and using multiple Availability Zones improves **high availability**.
- **Pricing and support plans** : **On-Demand** is for flexibility, **Reserved** and **Savings Plans** reduce cost for steady usage, **Spot** is cheapest for interruptible workloads, and higher support plans add faster help and more guidance.
- **Storage and databases** : **Amazon S3** is object storage, **Amazon EBS** is block storage for Amazon EC2, **Amazon EFS** is shared file storage, **Amazon RDS** is managed SQL, and **Amazon DynamoDB** is NoSQL.
- **Answer-style mini blocks** : **object storage = Amazon S3**, **managed relational database = Amazon RDS**, **NoSQL = Amazon DynamoDB**, **serverless code = AWS Lambda**, **audit API activity = AWS CloudTrail**, **metrics and alarms = Amazon CloudWatch**, **multiple account governance = AWS Organizations**, and **compliance reports = AWS Artifact**.

Common Question Patterns

- **Best service for a use case** : if the question says **object storage**, answer **Amazon S3**; if it says **managed relational database**, answer **Amazon RDS**; if it says **serverless code**, answer **AWS Lambda**.
- **Service comparisons** : **Amazon SNS** pushes messages to many subscribers, **Amazon SQS** stores messages in a queue, **AWS CloudTrail** records API activity, and **Amazon CloudWatch** monitors metrics, logs, and alarms.
- **Difference-focused exam notes** : **Amazon S3** stores objects, **Amazon EBS** provides block storage for Amazon EC2, and **Amazon EFS** provides shared file storage; **Amazon RDS** is SQL and **Amazon DynamoDB** is NoSQL; **AWS WAF** filters web requests and **AWS Shield** protects against DDoS attacks; **AWS Organizations** manages accounts and **AWS Control Tower** sets up and governs a multi-account environment.
- **Cost optimization choices** : for **steady workloads**, choose **Reserved Instances** or **Savings Plans**; for **flexible short-term workloads**, choose **On-Demand**; for **interruptible workloads**, choose **Spot Instances**.
- **Security responsibility** : **AWS** manages physical infrastructure, while the **customer** manages users, permissions, operating systems on Amazon EC2, application settings, and data protections.
- **Support and governance** : **AWS Trusted Advisor** gives recommendations, **AWS Artifact** provides compliance reports, **AWS Organizations** manages multiple accounts, and higher **AWS Support plans** add expert help and operational guidance.

SERVICE CATALOG

Official In-Scope AWS Services (Service Names Link to Official AWS Pages)

The categories below follow the official CLF-C02 in-scope services page. Descriptions are concise study notes written in official AWS style so the page stays readable. Icons come from the official AWS Architecture Icons package when AWS provides a dedicated icon for that service or tool.

Analytics

Amazon Athena

Interactive query service for analyzing data in Amazon Simple Storage Service (Amazon S3) with SQL.

Example: run SQL queries directly on log files stored in Amazon S3.

Amazon EMR

Cloud big data platform for processing large datasets with frameworks such as Apache Spark and Hadoop.

Example: process clickstream data with Apache Spark clusters.

AWS Glue

Serverless data integration service for discovering, preparing, moving, and combining data.

Example: prepare raw CSV data before loading it into a warehouse.

Amazon Kinesis

Streaming service for collecting, processing, and analyzing real-time data.

Example: capture sensor data from devices in real time.

Application Integration

Amazon EventBridge

Serverless event bus service for routing events between applications and AWS services.

Example: trigger a workflow when an Amazon EC2 instance changes state.

Amazon Simple Notification Service (Amazon SNS)

Pub/sub messaging service for fan-out notifications and event delivery.

Example: send one alarm to email, SMS, and another system at the same time.

Amazon Simple Queue Service (Amazon SQS)

Message queue service for decoupling distributed application components.

Example: queue orders before downstream processing.

Amazon OpenSearch Service

Managed search service for search, log analytics, and observability use cases.

Example: search application logs to troubleshoot production errors.

Amazon QuickSight

Business intelligence service for dashboards, analytics, and visual insights.

Example: create a dashboard for monthly sales performance.

Amazon Redshift

Data warehouse service designed for large-scale analytics.

Example: analyze years of reporting data across business units.

AWS Step Functions

Workflow service for coordinating application components and business processes.

Example: coordinate order validation, payment, and shipment steps.

Business Applications

Amazon Connect

Cloud contact center service for customer engagement.

Example: run a customer support call center in the cloud.

Amazon Simple Email Service (Amazon SES)

Email service for sending and receiving messages at scale.

Example: send password reset emails from an application.

Cloud Financial Management

AWS Budgets

Budgeting tool for setting custom cost and usage thresholds with alerts.

Example: get an alert when monthly cost goes above a target amount.

AWS Cost and Usage Reports

Detailed billing and usage reports for analysis and chargeback workflows.

Example: export granular billing data for finance analysis.

AWS Cost Explorer

Cost analysis tool for visualizing usage and spend trends.

Example: compare last month's Amazon EC2 cost with this month's.

AWS Marketplace

Digital catalog for finding, testing, and buying third-party software and services.

Example: subscribe to a third-party security product.

Compute

AWS Batch

Batch computing service for running large numbers of jobs.

Example: process nightly rendering or scientific jobs.

Amazon Elastic Compute Cloud (Amazon EC2)

Secure and resizable compute capacity in the cloud.

Example: run a Linux web server with full operating system control.

AWS Elastic Beanstalk

Application deployment service for quickly deploying and scaling web applications.

Example: deploy a web app without manually building all supporting infrastructure.

Amazon Lightsail

Simplified virtual private server service for websites, apps, and development environments.

Example: launch a small WordPress website quickly.

Containers

Amazon Elastic Container Registry (Amazon ECR)

Container image registry for storing, managing, and deploying container images.

Example: store Docker images before deployment.

Amazon Elastic Container Service (Amazon ECS)

Container orchestration service for running and scaling containerized applications.

Example: run microservices in containers without managing Kubernetes.

Amazon Elastic Kubernetes Service (Amazon EKS)

Managed Kubernetes service for running Kubernetes on AWS.

Example: operate Kubernetes workloads for a production platform.

AWS Outposts

AWS infrastructure on premises for hybrid deployments.

Example: run AWS services in a local data center for low-latency needs.

Customer Enablement

AWS Support

Technical support service with plan-based access to help and guidance.

Example: open a support case for a production issue.

Database

Amazon Aurora SQL

Relational database built for performance and availability in the cloud.

Example: run a high-availability MySQL-compatible production database.

Amazon DocumentDB NoSQL

Document database service with MongoDB compatibility.

Example: store product catalog documents in JSON-like form.

Amazon DynamoDB NoSQL

Serverless NoSQL database for key-value and document workloads.

Example: store shopping cart sessions at scale.

Amazon ElastiCache In-memory

In-memory caching service for accelerating applications.

Example: cache frequent database reads to reduce latency.

Amazon Neptune Graph

Graph database service for highly connected data.

Example: model social relationships or recommendation links.

Amazon Relational Database Service (Amazon RDS) SQL

Managed relational database service that simplifies setup, operation, and scaling.

Example: run PostgreSQL without handling backups manually.

Developer Tools

AWS Command Line Interface (AWS CLI)

Command line tool for interacting with AWS services.

Example: list Amazon S3 buckets from a terminal.

AWS CodeBuild

Build service for compiling source code, running tests, and producing software packages.

Example: run tests automatically after every code push.

AWS CodePipeline

Continuous delivery service for automating release pipelines.

Example: automate build, test, and deployment stages.

AWS X-Ray

Distributed tracing service for analyzing applications and requests.

Example: trace a slow request across microservices.

End User Computing

Amazon AppStream 2.0

Application streaming service for delivering desktop applications to users.

Example: stream a design application to remote users.

Amazon WorkSpaces

Managed virtual desktop service.

Example: give employees cloud desktops instead of physical PCs.

Amazon WorkSpaces Secure Browser

Secure browser service for protected web access.

Example: let contractors use only browser-based access to internal tools.

Frontend Web and Mobile

AWS Amplify

Development platform for building and hosting full-stack web and mobile applications.

Example: host a front end with authentication and API integration.

AWS AppSync

Managed GraphQL service with real-time and offline capabilities.

Example: power a mobile app backend with a GraphQL API.

Internet of Things (IoT)

AWS IoT Core

Managed device connectivity service for connecting and managing Internet of Things devices securely at scale.

Example: connect factory sensors to AWS securely.

Machine Learning

Amazon Comprehend

Natural language processing service for extracting insights from text.

Example: detect sentiment in customer reviews.

Amazon Kendra

Intelligent search service for enterprise content.

Example: search across company documents and FAQs.

Amazon Lex

Conversational interface service for chatbots and voice assistants.

Example: build a chatbot for booking appointments.

Amazon Polly

Text-to-speech service that turns text into lifelike speech.

Example: convert article text into spoken audio.

Amazon Q

Generative artificial intelligence assistant for AWS and business use cases.

Example: ask natural-language questions about AWS tasks.

Management and Governance

AWS Auto Scaling

Scaling service that automatically adjusts capacity to maintain performance and cost efficiency.

Example: add more Amazon EC2 instances when traffic rises.

AWS CloudFormation

Infrastructure as code service for modeling and provisioning AWS resources.

Example: create a network stack from a template.

AWS CloudTrail

Governance and audit service that records account activity and application programming interface (API) actions.

Example: identify who deleted a security group.

Amazon CloudWatch

Monitoring service for metrics, logs, alarms, and events.

Example: create an alarm for high CPU usage.

Amazon Rekognition

Image and video analysis service.

Example: detect objects in uploaded pictures.

Amazon SageMaker AI

Machine learning service for building, training, and deploying models.

Example: train a churn prediction model.

Amazon Textract

Document text extraction service for forms, tables, and scanned content.

Example: extract data from scanned invoices.

Amazon Transcribe

Automatic speech recognition service that converts speech to text.

Example: turn a call recording into searchable text.

Amazon Translate

Neural machine translation service.

Example: translate product descriptions into another language.

AWS Compute Optimizer

Optimization service that recommends right-sized AWS resources.

Example: get advice to move to a smaller instance size.

AWS Config

Configuration tracking service for resource inventory, history, and compliance.

Example: detect whether a bucket became publicly accessible.

AWS Control Tower

Multi-account governance service for setting up and managing AWS environments.

Example: apply guardrails to a new organization setup.

AWS Health Dashboard

Health visibility service for account-specific and service events.

Example: see whether an AWS issue affects your account.

AWS License Manager

License management service for tracking and controlling software licenses.

Example: manage Microsoft license usage on compute resources.

AWS Management Console

Web console for managing AWS resources in a browser.

Example: create an Amazon S3 bucket from the console.

AWS Organizations

Account management service for centrally governing multiple AWS accounts.

Example: manage billing across several company accounts.

AWS Service Catalog

Catalog management service for approved cloud products and templates.

Example: let teams launch only approved infrastructure products.

Service Quotas

Quota management feature for viewing and requesting service limit increases.

Example: request a higher Amazon EC2 vCPU limit.

AWS Systems Manager

Operations management service for visibility and control of AWS resources.

Example: patch a fleet of servers centrally.

AWS Trusted Advisor

Recommendation service for cost optimization, security, performance, and resilience.

Example: identify idle resources or open security groups.

AWS Well-Architected Tool

Architecture review service for evaluating workloads against AWS best practices.

Example: review a workload for reliability and cost optimization.

Migration and Transfer

AWS Application Discovery Service

Discovery service for collecting data from on-premises servers for migration planning.

Example: inventory existing servers before moving workloads.

AWS Application Migration Service

Lift-and-shift migration service for rehosting servers on AWS.

Example: move an on-premises Windows server to AWS.

AWS Database Migration Service (AWS DMS)

Database migration service for moving databases with minimal downtime.

Example: migrate a production database to Amazon RDS.

Migration Evaluator

Migration business-case assessment tool.

Example: estimate savings before migrating from on-premises infrastructure.

Networking and Content Delivery

Amazon API Gateway

API management service for creating, publishing, and securing APIs.

Example: expose a backend service to a mobile app through an API.

Amazon CloudFront

Content delivery network service for low-latency global delivery.

Example: cache website assets closer to users worldwide.

AWS Direct Connect

Dedicated network connection service to AWS.

Example: connect a corporate data center directly to AWS.

AWS Global Accelerator

Network acceleration service for improving global application availability and performance.

Example: improve performance for users accessing a global app.

AWS Migration Hub

Migration tracking service for monitoring application migrations.

Example: track progress across multiple migration projects.

AWS Schema Conversion Tool (AWS SCT)

Schema conversion tool for heterogeneous database migrations.

Example: convert a schema from SQL Server to PostgreSQL.

AWS Snow Family

Physical edge and transfer devices for moving data in and out of AWS.

Example: transfer large datasets when network bandwidth is limited.

AWS PrivateLink

Private connectivity service for accessing services without public internet exposure.

Example: privately access a service from within a virtual private cloud.

Amazon Route 53

Domain Name System web service for routing end users to applications.

Example: direct traffic from a domain name to a load balancer.

AWS Transit Gateway

Network transit hub for connecting virtual private clouds (VPCs) and on-premises networks.

Example: connect many VPCs through one central network hub.

Amazon Virtual Private Cloud (Amazon VPC)

Isolated virtual network for launching AWS resources.

Example: place databases in private subnets and web servers in public ones.

AWS Virtual Private Network (AWS VPN)

Encrypted network connection service between AWS and external networks.

Example: create secure connectivity between your office and AWS.

AWS Site-to-Site VPN

Site connection service for linking on-premises networks to AWS.

Example: connect a branch office network to AWS.

AWS Client VPN

Remote access virtual private network service for end users.

Example: let remote employees securely access internal AWS resources.

Security, Identity, and Compliance

AWS Artifact

Compliance access service for on-demand AWS security and compliance reports.

Example: download compliance documents for an audit.

AWS Audit Manager

Audit evidence service for collecting and organizing evidence.

Example: gather evidence for an internal compliance review.

AWS Certificate Manager (ACM)

Certificate management service for Secure Sockets Layer / Transport Layer Security (SSL/TLS) certificates.

Example: attach an SSL/TLS certificate to a load balancer.

AWS CloudHSM

Hardware security module service for generating and using encryption keys.

Example: keep encryption keys in dedicated hardware.

Serverless

AWS Fargate

Serverless compute engine for containers.

Example: run containers without provisioning servers.

AWS Lambda

Serverless compute service for running code in response to events.

Example: trigger code when a file is uploaded to storage.

Amazon Cognito

User identity service for sign-up, sign-in, and access control in applications.

Example: let users sign in to a mobile application.

Amazon Detective

Security investigation service for analyzing findings and causes.

Example: investigate unusual activity flagged by another security service.

AWS Directory Service

Managed directory service for directory-aware workloads.

Example: connect AWS workloads to Microsoft Active Directory.

AWS Firewall Manager

Centralized firewall policy management service across accounts.

Example: apply the same web protection rules in many accounts.

Amazon GuardDuty

Threat detection service for continuous monitoring and malicious activity detection.

Example: detect suspicious API activity in an account.

AWS Identity and Access Management (AWS IAM)

Access management service for users, roles, and permissions.

Example: create a role for a service to read from a bucket.

AWS IAM Identity Center

Centralized workforce access service for AWS accounts and applications.

Example: let employees sign in once to multiple AWS accounts.

Amazon Inspector

Vulnerability management service for automated security assessments.

Example: scan compute resources for vulnerabilities.

AWS Key Management Service (AWS KMS)

Encryption key management service.

Example: encrypt stored data with customer-managed keys.

Amazon Macie

Sensitive data discovery service for Amazon Simple Storage Service (Amazon S3).

Example: find exposed personal data in storage buckets.

AWS Resource Access Manager (AWS RAM)

Resource sharing service across AWS accounts.

Example: share a network resource across accounts.

AWS Secrets Manager

Secret storage service for credentials and application secrets.

Example: rotate a database password automatically.

AWS Security Hub

Security posture management service that centralizes findings and checks.

Example: review security findings from multiple services in one place.

AWS Shield

Distributed denial of service protection service.

Example: protect a public-facing application from volumetric attacks.

AWS Web Application Firewall (AWS WAF)

Web traffic filtering service for protecting web applications.

Example: block common web exploits against an API.

Storage

AWS Backup

Backup management service for centralizing and automating backups.

Example: back up multiple services with one central policy.

Amazon Elastic Block Store (Amazon EBS)

Block storage service for use with Amazon Elastic Compute Cloud (Amazon EC2).

Example: attach persistent storage to a compute instance.

Amazon Elastic File System (Amazon EFS)

File storage service for scalable shared access.

Example: share files across multiple application servers.

AWS Elastic Disaster Recovery

Disaster recovery service for rapid recovery of physical, virtual, and cloud-based servers.

Example: recover important servers after a site outage.

Amazon FSx

Managed file system service for popular file workloads.

Example: run a managed Windows file share.

Amazon Simple Storage Service (Amazon S3)

Object storage service built to store and retrieve any amount of data.

Example: store website assets, backups, or logs.

Amazon S3 Glacier

Archive storage option for low-cost long-term data retention.

Example: retain compliance records for years at low cost.

AWS Storage Gateway

Hybrid storage service for connecting on-premises environments with AWS storage.

Example: link an on-premises backup application to AWS storage.

Domain-Referenced AWS Tools and Additional Services

These items appear in the official CLF-C02 domain pages, but they are not listed in the official in-scope services table above. Official AWS Architecture Icons are shown where AWS provides a matching icon in the current package.

AWS Snowball

Data transfer device used to move large amounts of data into and out of AWS when network transfer is limited or impractical.

Example: ship large backup datasets to AWS instead of uploading them over a slow connection.

AWS Pricing Calculator

Cost estimation tool used to model AWS architectures and estimate expected monthly spend before deployment.

Example: estimate the cost of running a web app with Amazon EC2, Amazon RDS, and Amazon S3.

AWS Health API

Programmatic health access for retrieving AWS Health events and account-specific service notifications through code or automation.

Example: pull AWS Health events into an internal operations dashboard.

REFERENCE

Official AWS Sources

These are the primary official AWS pages used to structure this revision sheet.

- CLF-C02 exam guide
- AWS Skill Builder learning plan
- Content Domain 1: Cloud Concepts
- Content Domain 2: Security and Compliance
- Content Domain 3: Cloud Technology and Services
- Content Domain 4: Billing, Pricing, and Support
- Technologies and Concepts
- In-Scope AWS Services
- AWS Architecture Icons